Case Study Part 3:

Tesla and the Electric Vehicle Revolution: Analysis of Product Development

By

Naw Mu Aye, Naw Mu Aye

Olatowoju, Joshua

San Miguel, Roberto

Vargas Gomez, Juan Jose

An assignment submitted to

Dr. Karim Souidi

Faculty of Master's in Data Analytics

In Fulfillment of the Requirements for the Course

DAMO-511-1: Data Analytics Case of Study 2

At

University of Niagara Falls Canada

March 2025

Introduction

This report aims to analyze Tesla's development and market performance through the PESTEL framework, evaluating Political, Economic, Social, Technological, Environmental, and Legal factors that shape the company's growth. By using data dashboards, this study provides insights into how macroeconomic trends, regulatory policies, technological advancements, and consumer behavior influence Tesla's strategic decision-making. The analysis identifies both challenges and opportunities, giving some glance on how Tesla can navigate market uncertainties while maintaining its competitive edge. Ultimately, this study serves as a comprehensive evaluation of Tesla's position in the electric vehicle (EV) industry and its potential for future expansion.

PESTEL Factor Analysis

Political Factors

Political factors play a crucial role in shaping Tesla's growth in the U.S. electric vehicle (EV) market. Government policies, trade relations, and regulatory changes have directly impacted Tesla's financial performance, stock prices, and sales revenue. The following analysis examines the political influences on Tesla, based on *Figure 1* and key factors.

1. U.S.-China Trade Tensions and Domestic EV Production

The ongoing trade tensions between the U.S. and China have influenced Tesla's market strategy. With tariffs and restrictions on Chinese imports, Tesla has benefited from policies favoring domestic EV production. The graphs indicate a steady rise in Tesla's sales revenue and stock price, suggesting that political pressure to reduce dependence on foreign supply chains has encouraged domestic EV adoption.

Key Insights: The trade tensions have had a positive impact on Tesla by encouraging domestic EV production. Tariffs and restrictions on Chinese imports have pushed Tesla to

focus more on local manufacturing. This has led to a steady rise in both Tesla's sales revenue and stock price, indicating that domestic policy shifts favoring local production have helped the company thrive.

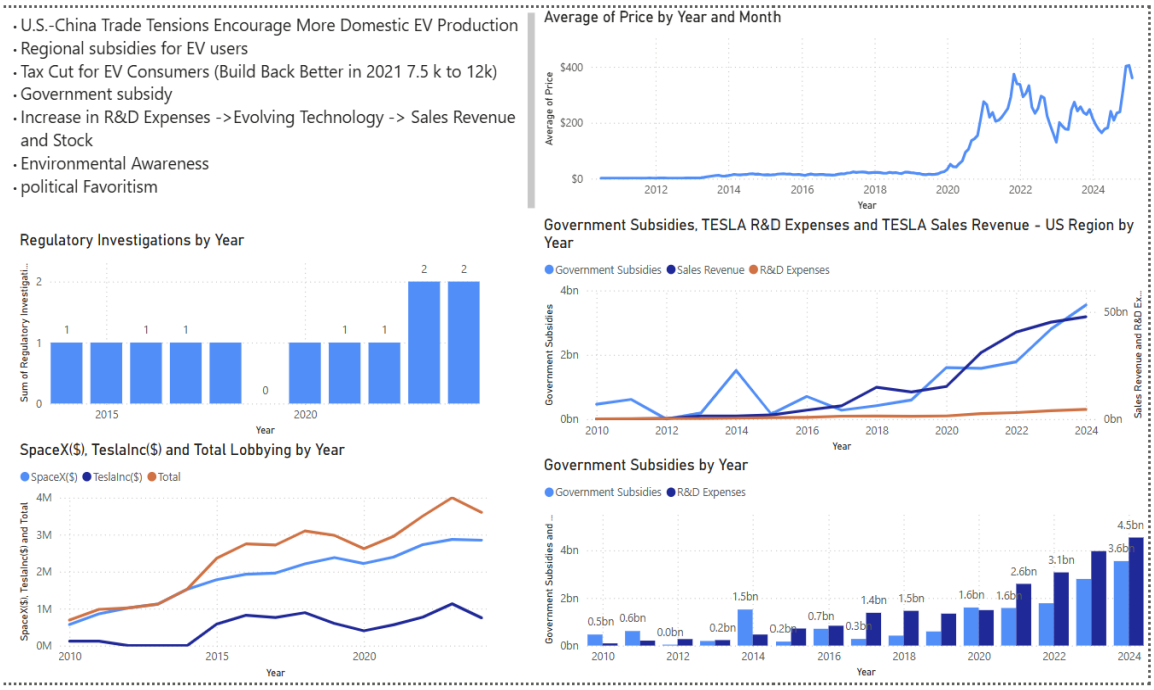

Figure 1: Political Factor Impact Analysis on Tesla EV sector

2. Government Subsidies and R&D Investments

Tesla has received substantial government subsidies, as shown in the "Government Subsidies by Year" and "Government Subsidies, R&D Expenses, and Sales Revenue" charts in *Figure 1*. These subsidies have helped Tesla increase its R&D spending, leading to technological advancements and higher production capacity. The correlation between rising R&D expenses and sales revenue demonstrates how financial support from the government has fueled Tesla's expansion.

Key Insights: Government subsidies have been a significant financial support for Tesla. These subsidies have allowed Tesla to ramp up its R&D spending, which, in turn, has driven technological innovation and expanded production capacity. The data shows a direct

correlation between higher R&D investments and increasing sales revenue, highlighting the importance of financial support for Tesla's growth and development.

3. Tax Incentives for EV Consumers

The Build Back Better Act (2021) increased tax credits for EV buyers from $7,500 to $12,000, making Tesla vehicles more affordable and boosting demand (Wikipedia, 2025) (Peterson Institute for International Economics, 2022). Despite these tax incentives remaining relatively stable over time, their impact has been significant in encouraging EV adoption. The graphs in *Figure 1* show a clear upward trend in sales revenue following major tax policy changes.

Key Insights: The increase in tax credits for EV buyers, especially through initiatives like the Build Back Better Act, has played a pivotal role in making Tesla vehicles more affordable. The graphs show a clear upward trend in sales revenue following tax policy changes, underlining the effectiveness of such incentives in promoting EV adoption and boosting consumer demand for Tesla cars.

4. Political Favoritism and Lobbying Efforts

Tesla's lobbying efforts have steadily increased, as reflected in the "SpaceX, Tesla Inc., and Total Lobbying by Year" chart OpenSecrets. (n.d.). By actively engaging with policymakers, Tesla has influenced regulations and secured favorable policies. This suggests that political favoritism has played a role in Tesla's ability to secure subsidies, avoid excessive regulations, and maintain a competitive edge in the EV sector.

Key Insights: Tesla's active lobbying efforts have helped the company secure favorable policies and subsidies while also influencing regulations. The increase in lobbying activities, as shown in the data, suggests that Tesla's political engagement has been successful in ensuring advantageous conditions for its operations, giving it a competitive edge in the EV sector.

5. Regulatory Environment and Compliance

The **"Regulatory Investigations by Year"** chart shows fluctuations in the number of investigations Tesla has faced. While Tesla has generally complied with emission and EV regulations, occasional legal and regulatory challenges have affected its operations. Despite this, the overall political environment has remained supportive of Tesla's growth.

Key Insights: Tesla has faced occasional regulatory investigations, as indicated by fluctuations in the number of investigations over the years. While these challenges have impacted Tesla's operations at times, the overall regulatory environment has been supportive of Tesla's growth, with Tesla generally adhering to emission and EV regulations.

6. Overall Insights

Political factors, including government subsidies, tax incentives, trade policies, and lobbying efforts, have significantly influenced Tesla's trajectory. The company has strategically leveraged political dynamics to expand its production, increase R&D investment, and boost EV adoption. Moving forward, Tesla's success will continue to depend on how well it navigates policy changes, regulatory challenges, and geopolitical tensions.

Economic Factors

Economic factors play a crucial role in Tesla's growth, influencing its workforce expansion, market share, sales revenue, and investment in research and development. The graphs provide key insights into how Tesla's performance aligns with macroeconomic trends in the U.S., particularly GDP growth, government incentives, and overall EV market dynamics.

1. US GDP and Tesla's Workforce Size

As illustrated in *Figure 2* US GDP and Tesla Workforce by Year plot, Tesla's workforce has steadily increased, particularly after 2020, aligning with a rise in U.S. GDP.

Economic growth generally fosters increased demand for electric vehicles (EVs), leading to job creation within Tesla. A drop in GDP during economic downturns (e.g., COVID-19 in 2020) appears to correlate with a temporary dip in Tesla's workforce expansion.

Key Insights: Tesla's workforce expansion aligns with U.S. GDP growth, particularly after 2020. Economic growth drives increased demand for electric vehicles (EVs), which leads to job creation within Tesla. However, during economic downturns (e.g., the 2020 COVID-19 recession), a temporary dip in GDP correlates with a slowdown in workforce expansion. Tesla benefits from economic expansion but remains sensitive to downturns that could affect hiring and operational capacity.

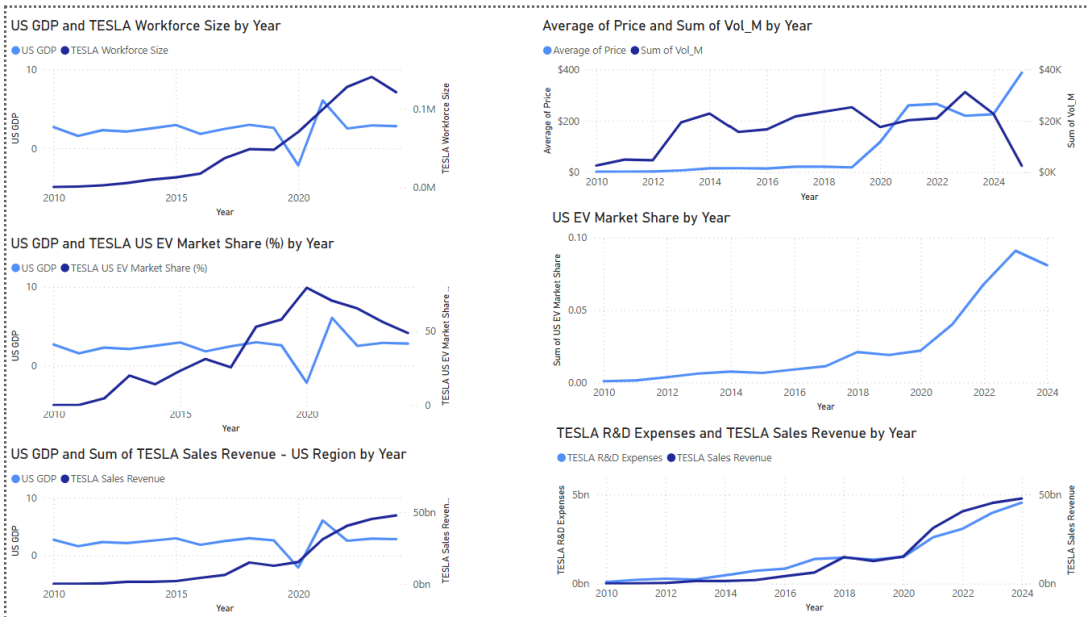

Figure 2: Economical Factor Analysis on Tesla EV Sector

2. US GDP and Tesla's EV Market Share

As illustrated in *Figure 2* US GDP and Tesla US EV Market Share by Year plot, Tesla's U.S. EV market share saw significant growth post-2015, peaking around 2021-2022. The correlation between GDP and Tesla's market share suggests that economic stability supports EV adoption.

The recent decline in market share may indicate increased competition from other automakers entering the EV sector.

Key Insights: Tesla's U.S. EV market share grew significantly after 2015, peaking in 2021-2022. This growth is closely tied to economic stability, as a stronger GDP typically supports the adoption of EVs. However, the recent decline in Tesla's market share suggests that increased competition from other automakers has begun to affect its dominance in the EV market. While Tesla continues to lead, it faces challenges from both macroeconomic factors and rising competition.

3. Tesla's Stock Price and Trading Volume Trends

Tesla's stock price has shown strong growth, with major surges post-2020 according to the Average of Stock Price and Sum of Trade Volume by Yar plot in *Figure 2*.

Trading volume fluctuations suggest that economic events, investor confidence, and policy shifts drive stock activity. Economic uncertainty (e.g., trade policies, interest rates) often trigger shifts in stock volume.

Key Insights: Tesla's stock price has shown strong growth, particularly post-2020, with notable surges reflecting investor confidence. Trading volume fluctuations suggest that Tesla's stock is highly sensitive to economic events and shifts in investor sentiment, such as changes in trade policies and interest rates. Economic uncertainty can significantly influence stock activity, indicating that Tesla's stock price is reactive to broader economic conditions.

4. US GDP and Tesla's Sales Revenue (US Region)

Tesla's U.S. sales revenue has grown significantly, reflecting strong EV adoption amid GDP growth. Government subsidies and tax incentives have played a role in sustaining revenue growth, even during periods of economic instability.

A decline in GDP in 2020 did not significantly impact Tesla's revenue, possibly due to increased consumer preference for EVs and continued government support.

Key Insights: Tesla's U.S. sales revenue has grown significantly, reflecting strong EV adoption, which is supported by GDP growth. Despite economic instability, including the 2020 GDP decline, Tesla's revenue remained resilient due to consumer preference for EVs and continued government subsidies and tax incentives. Tesla's revenue is robust, but still vulnerable to broader economic trends, highlighting the importance of macroeconomic stability for sustaining revenue growth.

5. Tesla's R&D Expenses and Sales Revenue

Tesla's increasing investment in R&D has paralleled rising sales revenue. Higher R&D spending likely drives technological advancements, leading to higher consumer demand and sales. The steady rise in both metrics suggests that innovation remains a critical driver of Tesla's financial performance.

Key Insights: Tesla's increasing investment in R&D is closely linked to rising sales revenue. Higher R&D spending drives technological innovations that lead to increased consumer demand, contributing to sales growth. The consistent rise in both metrics shows that innovation is a critical driver of Tesla's financial performance. Continued investment in R&D is essential for long-term growth, although economic conditions and access to funding could influence the pace of innovation.

6. Overall Insights

Tesla's performance is significantly influenced by macroeconomic trends, including U.S. GDP growth, competition, investor sentiment, and government policies. While the company has been resilient in the face of economic fluctuations, its ability to maintain growth will depend on continued economic stability, effective competition management, and strategic investment in R&D. The interplay between economic factors and Tesla's business strategies will shape its future growth prospects.

Social Factors

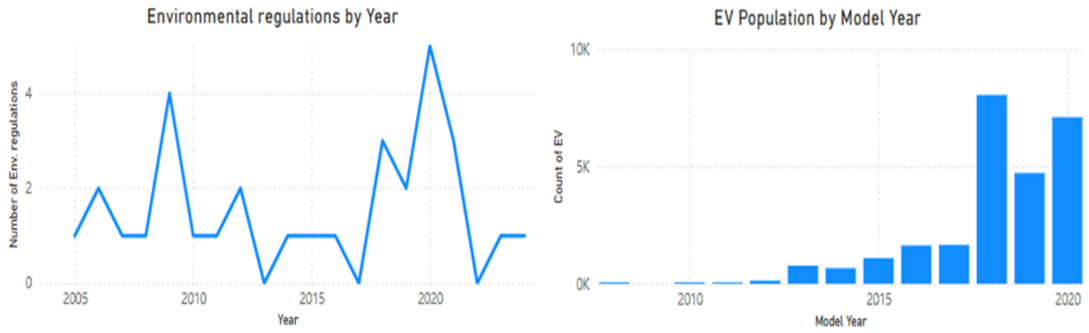

Figure 3: *Social Factors Dashboard*

1. EV Adoption

The increasing number of EVs over the years highlights a social trending towards sustainable transportation, mostly driven by environmental awareness and government incentives, as we can interpret from *Figure 3*.

Key Insights: The barrier broken in 2018 which was one year after the launch of Model X shows that, with the right vehicle, people are open to embrace electric vehicles instead of those working with gas.

2. Consumer Preferences

The division between SUVs and sports cars plays a role in purchasing behavior. SUVs (Model X, Model Y) appeal to practical-minded consumers, while sports models (Model S, Model 3, Roadster) attract those seeking performance and luxury. This division can be seen when we check that SUVs model's launch were around 2016 to 2018.

Key Insights: Consumers priority isn't longer ranges on Tesla's offers, but rather luxury. Despite that, as luxury increases, it is expected to see an increase on the electric range of the vehicles.

3. Luxury Market Positioning

Tesla's vehicles are perceived as premium products, which influences consumer demographics and market segmentation (Rezzano, Fallini, & Brun, 2021).

Key Insights: Tesla's branding as a luxury EV manufacturer positions it favourably in the high-end market but may limit accessibility to price-sensitive consumers. Although making the right fixes this luxury barriers can be weakened for a broader audience to take interest on Tesla's offers.

4. Overall Insights

The connection between social and technological factors has been fundamental in shaping Tesla's product development strategy. As Tesla's image gained popularity and strength in the USA, the expansion into the SUV category demonstrates a strategic understanding of the market. By integrating luxury vehicles, traditionally associated with the sports segment, Tesla managed to reach a broader audience. This decision to offer luxurious yet practical models was key to their sustained growth over the years.

Technological Factors

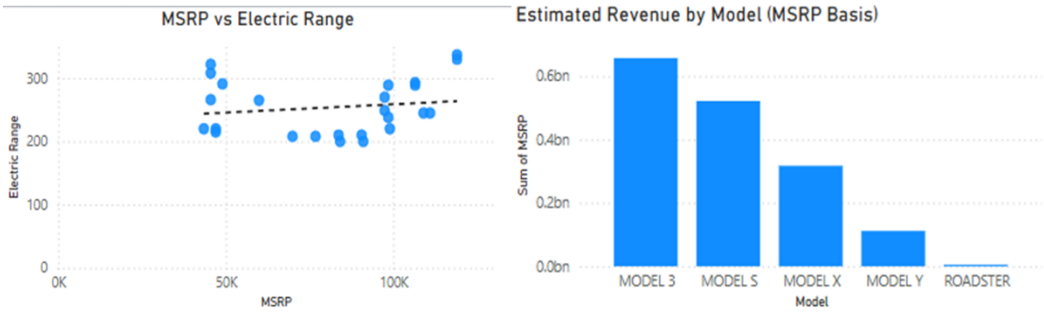

Figure 4:Technological factors

1. MSRP vs. Electric Range Analysis

The data (*Figure 4*) reveals a U-shaped trend, showing that price and range are not directly proportional. Instead, they are influenced by technological optimization and design priorities.

Key Insights: The U-shaped trend in MSRP vs. Electric Range suggests that Tesla does not price vehicles solely based on battery range. Instead, pricing reflects vehicle type,

technology optimization, and performance priorities. Luxury models may focus on acceleration, premium materials, and advanced features instead of pure range extension.

2. Impact of Model Year in Performance vs. Range Trade-off

When analyzing by model year, there is no clear correlation between newer models and increased electric range. Some Tesla models prioritize power, speed, and premium amenities over maximizing range per charge, meaning not all technological improvements focus on extending battery life

Key Insights: As the automobile industry works, companies usually create a model of a car, based on that model they made iterations where more features are added to the vehicles. In Tesla's case, seems to implicate that people is fine by having a good electric range performance but are more interested in what these cars can offer, like intelligent panels and other features Tesla has been adding to their products.

3. Battery Efficiency & Energy Consumption

Optimization in battery technology affects vehicle range, but higher-priced models do not necessarily achieve the longest range. Instead, manufacturers balance performance, efficiency, and luxury features.

Key Insights: Tesla's technology is designed not just to increase range, but to optimize efficiency and performance based on the intended use of each model. As a basic rule, SUVs may have higher battery consumption due to weight, while sports cars may sacrifice range for speed and acceleration, which is an important feature for luxury-sport cars.

4. Overall Insights

Tesla's technological advancements prioritize an approach to efficiency and performance, focusing on optimized battery use rather than simply extending range linearly. This is evident in the lack of a direct correlation between model year and increased range, indicating Tesla's strategic balancing of cost, efficiency, and luxury features to cover diverse

consumer needs across different vehicle types. Rather than pursuing uniform range improvements across all models, Tesla manage its strategy to each market segment. Looking forward, future advancements should concentrate on enhancing battery energy density while maintaining the delicate balance of performance and affordability, ensuring sustained innovation and market competitiveness.

Environmental Factors

Environmental factors directly influence Tesla's vehicle efficiency, battery technology, and emissions reduction strategies. The dashboard presented on *Figure 5* provides insights into Tesla's sustainability efforts and challenges in meeting global environmental standards.

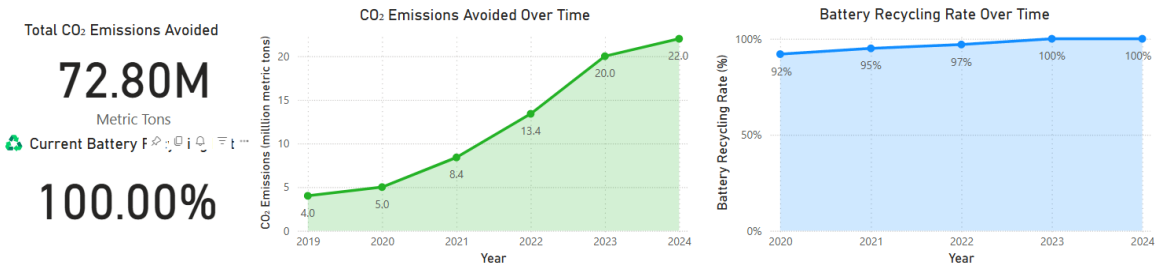

Figure 5:Environmental Factors Dashboard

1. CO_2 Emissions Avoided Over Time

Tesla's commitment to reducing CO_2 emissions is evident from the Power BI visualization, which shows a steady rise from 4.0 million metric tons (2019) to 22 million metric tons (2024). This increase is attributed to higher EV adoption, efficiency improvements, and expanded Gigafactory production. Tesla's ability to reduce carbon emissions aligns with global regulatory policies, including the EU Green Deal and the U.S. Inflation Reduction Act.

Key Insights: Tesla's CO_2 reduction efforts enhance its sustainability-driven brand image, attracting ESG investors and securing regulatory incentives. However, maintaining emission reduction targets requires continuous battery efficiency improvements.

2. Battery Recycling Rate Over Time

The dashboard data shows Tesla's battery recycling rate increased from 92% (2020) to 100% (2023). This achievement ensures supply chain sustainability and reduces dependence on newly mined raw materials. Tesla's closed-loop recycling model enables recovery of lithium, cobalt, and nickel, aligning with the EU Battery Directive.

Key Insights: Higher recycling rates reduce Tesla's reliance on external suppliers, supporting cost-efficient battery innovation. However, Tesla still depends on third-party material extraction, posing long-term supply chain risks.

3. Overall Insights

Tesla significantly reduces CO_2 emissions and advances battery recycling, solidifying its sustainability leadership. Increased EV adoption and efficient production, driven by global regulations like the EU Green Deal and the U.S. Inflation Reduction Act, directly result in rising emissions avoidance. Tesla's closed-loop recycling further reduces reliance on raw materials, supporting innovation and compliance. While these efforts boost brand value and attract investment, long-term sustainability hinges on continuous battery improvements and reduced dependence on external material extraction to secure supply chains.

Legal Factors

Legal and regulatory factors impact Tesla's compliance with vehicle safety standards, labour laws, and cybersecurity protocols. The dashboard presented in *Figure 6* visualizes trends in regulatory investigations and cybersecurity compliance, providing insights into Tesla's legal challenges.

Figure 6: Legal Factors Dashboard

1. Regulatory Investigations Over Time

Tesla's regulatory investigations doubled from 1 case per year (2014-2018) to 2 per year in 2023-2024. Most cases focus on Autopilot safety, labor policies, and emissions compliance. Government agencies, including the NHTSA and the EU Commission, have increased scrutiny over Tesla's self-driving technology, affecting Autopilot software development and deployment.

Key Insights: Increasing regulatory pressure delays Tesla's full self-driving (FSD) feature enhancements. Tesla must prioritize transparency and compliance improvements to accelerate product approvals.

2. Cybersecurity Compliance Over Time

Tesla's cybersecurity compliance rate is 54.55%, exposing vulnerabilities in AI-driven vehicle systems and data privacy protocols. Weak compliance raises concerns about hacking threats, software security breaches, and over-the-air (OTA) update risks.

Key Insights: Weak cybersecurity standards hinder Tesla's AI-driven product innovations. Strengthening vehicle encryption and enhancing compliance with GDPR and CCPA regulations are essential to sustaining Tesla's market trust.

3. Overall Insights

Tesla faces growing regulatory scrutiny and cybersecurity challenges, which impact its technological advancements and market trust. Regulatory investigations have doubled in

recent years, primarily targeting Autopilot safety, labour policies, and emissions compliance, leading to delays in Full Self-Driving (FSD) feature enhancements. Simultaneously, Tesla's cybersecurity compliance rate of 54.55% highlights vulnerabilities in AI-driven vehicle systems and data privacy, increasing risks of hacking and software breaches. To sustain its market leadership and regulatory approval processes, Tesla must strengthen compliance measures, enhance transparency, and improve cybersecurity protocols to mitigate risks and accelerate innovation.

Operational Risk & Opportunity

This section justifies why each variable is important to the analysis and how it relates to the operational focus of Tesla. The selected variables are categorized under PESTEL (Political, Economic, Social, Technological, Environmental, and Legal) factors, which influence Tesla's market performance, risk assessment, and strategic decision-making.

Political Factor

Operational Risks

1. **Regulatory Uncertainty**
   - Changes in **EV tax credits and government subsidies** could impact Tesla's pricing strategy and demand.
   - Political shifts may lead to **stricter emissions and data privacy laws**, increasing compliance costs.
2. **Geopolitical Tensions**
   - **U.S.-China trade conflicts** could disrupt Tesla's **supply chain** and increase production costs.
   - **Tariffs on Chinese-made batteries or raw materials** may raise Tesla's manufacturing expenses.
3. **Dependence on Government Incentives**

- A reduction in **EV subsidies** could slow down demand, affecting revenue.
- Policy changes favoring **competitors (e.g., subsidies for other automakers)** might reduce Tesla's market share.

4. **Regulatory Investigations and Compliance Risks**
   - Increased **scrutiny from government agencies** (e.g., labor practices, cybersecurity regulations) could lead to fines or reputational damage.
   - Delays in meeting **EV compliance regulations** might result in penalties.

Operational Opportunities

1. **Government Support for EV Expansion**
   - Ongoing federal and state incentives encourage **higher EV adoption**, boosting Tesla's sales.
   - **Infrastructure bills** supporting **EV charging stations** will increase demand for Tesla vehicles.
2. **R&D and Innovation Growth**
   - Government funding for **sustainable technology and AI-driven automation** could accelerate Tesla's **battery advancements and self-driving tech**.
   - Policies supporting **EV manufacturing in the U.S.** could help Tesla **expand production facilities**.
3. **Trade Policy Benefits**
   - Tariffs on foreign EV competitors could **increase Tesla's dominance in the U.S. market**.
   - **Government support for domestic battery production** may reduce costs and improve supply chain resilience.
4. **Lobbying and Political Influence**

- Tesla's increasing **lobbying efforts** give it leverage to **shape future EV regulations** in its favor.
- Favorable policies supporting **renewable energy and carbon credits** may create new revenue streams.

Economic Factor

Operational Risks

1. **Economic Downturns:** A slowdown in GDP growth or a recession could reduce EV demand and impact Tesla's workforce and revenue.
2. **Market Competition:** Increasing competition from legacy automakers and new EV startups may challenge Tesla's market share.
3. **Stock Market Volatility:** Tesla's stock is highly reactive to macroeconomic changes, leading to investor uncertainty.
4. **Dependence on Government Incentives:** A reduction in EV tax credits or subsidies could slow adoption and revenue growth.

Operational Opportunities

1. **Economic Growth:** A strong GDP outlook supports higher consumer purchasing power and EV adoption.
2. **Technology-Led Growth**: Increased R&D investment enhances Tesla's competitive edge and product innovation.
3. **Expansion Potential:** Tesla can capitalize on growing U.S. EV demand and infrastructure expansion.
4. **Sustainable Revenue Streams**: Diversification into battery technology and energy storage could mitigate risks from vehicle sales fluctuations.

Social Factor

Operational Risks

1. **Affordability Concerns:** High MSRP values could limit access to broader consumer segments, especially in price-sensitive markets.
2. **Market Competition:** Increasing EV competition from both legacy automakers and new entrants may impact Tesla's market share and innovation pace, even more when Tesla’s image is not only depending on its products or services.

Operational Opportunities

1. **Consumer Adoption:** The rising population of EVs indicates increasing consumer acceptance, driven by awareness of sustainability and government incentives.
2. **Regulatory and Social Media Support:** Growing environmental regulations encourage EV adoption, creating a favorable market landscape, while the social media shows how people are embracing more options different from vehicles working with gas.

Technological Factor

Operational Risks

1. **Battery Technology Challenges**: Dependence on lithium-ion batteries presents challenges such as supply chain constraints and raw material availability. Though, thanks to other analysis we know Tesla has these topics well covered. Other challenge is having to many features or amenities on each car which could drastically vary overall performance with newer technology implemented on each car model.

2. **Market Competition**: The increased popularity of EV through the years has driven the eyes of other companies towards this market, meaning that Tesla requires to keep researching how to improve performance while delivering luxury products.

Operational Opportunities

1. **Technological Advancements**: Continuous improvements in battery range enhance the appeal of Tesla's products, reinforcing its market leadership. Also increasing the efficiency of battery consumption by the amenities each model has.
2. **Different Sources for Electric Chargers:** The advances in technology investigation regarding sources for clean energy increases the chances to maintain and increase the relevance EV vehicles have, for example, by how fast this energy could be charged in every car.

Environmental Factor

Operational Risks

1. **Emission Reduction Pressure**: Tesla must sustain aggressive carbon reduction targets, requiring lighter and more energy-dense battery designs.
2. **Raw Material Dependency**: Despite achieving 100% battery recycling, Tesla relies on external lithium and cobalt suppliers, impacting long-term battery innovation.

Operational Opportunities

1. **Next-Gen Battery Innovation**: Sustainability policies drive Tesla to develop solid-state batteries with longer lifecycles.
2. **Regulatory Incentives:** Tesla benefits from government incentives for sustainable product development, supporting energy-efficient vehicle models.

Legal Factor

Operational Risks

1. **Autopilot Regulations**: Increased scrutiny on self-driving technology delays Tesla's FSD advancements.
2. **Cybersecurity Vulnerabilities**: Weak compliance exposes Tesla's AI-driven vehicles to data security risks, slowing innovation.

Operational Opportunities

1. **Stronger Compliance for Faster Product Approvals** Enhanced legal transparency accelerates Tesla's regulatory approvals for FSD systems.
2. **Cybersecurity Investments for AI Safety:** Improving vehicle encryption and data security ensures safer AI-driven software updates.

Strategic Recommendations for Tesla

1. Policy & Market Positioning

- Strengthen lobbying efforts to secure long-term EV subsidies and favorable policies.
- Expand domestic production to mitigate trade tensions and supply chain disruptions.

2. Financial & Economic Strategy

- Diversify revenue streams by expanding into energy storage and AI-driven vehicle services.
- Introduce more affordable EV models to capture a broader customer base

3. Innovation & Technology

- Accelerate R&D in next-gen battery technology (solid-state batteries) to reduce costs and improve efficiency.
- Strengthen cybersecurity measures to protect autonomous driving and user data.

4. Environmental & Sustainability Goals

- Enhance battery recycling and sustainable sourcing to reduce regulatory risks.
- Expand charging infrastructure to support long-term EV adoption.

5. Risk Management & Compliance

- Improve regulatory compliance to navigate legal challenges, especially in FSD technology.
- Strengthen risk mitigation strategies for stock volatility and competitive pressures.

By implementing these strategies, Tesla can sustain its leadership in the EV market while mitigating operational risks and maximizing growth opportunities.

Conclusion

Tesla operates in a complex and dynamic environment where political, economic, social, technological, environmental, and legal factors significantly impact its trajectory. The company has demonstrated resilience in economic fluctuations, leveraging government incentives, technological advancements, and consumer trends to sustain growth. However, increasing regulatory scrutiny, cybersecurity vulnerabilities, and intensified market competition present significant challenges. To maintain its leadership in the EV industry, Tesla must continue to innovate in battery technology, enhance cybersecurity measures, and effectively navigate regulatory landscapes. Additionally, balancing sustainability goals with supply chain resilience will be crucial for long-term success. By strategically addressing these factors, Tesla can reinforce its market position and drive the transition toward sustainable transportation on a global scale.

References

1. Wikipedia. (2025, February 25). *Build Back Better Act*. Retrieved February 7, 2025, from https://en.wikipedia.org/wiki/Build_Back_Better_Act

2. Peterson Institute for International Economics. (2022, August 29). *Why US allies are upset over electric vehicle subsidies in the Inflation Reduction Act*

. Retrieved February 7, 2025, from https://www.piie.com/blogs/realtime-economics/2022/why-us-allies-are-upset-over-electric-vehicle-subsidies-inflation

3. OpenSecrets. (n.d.). *Tesla Inc lobbying profile*. OpenSecrets. https://www.opensecrets.org/federal-lobbying/clients/summary?id=D000057516

4. OpenSecrets. (n.d.). *SpaceX lobbying profile*. https://www.opensecrets.org/federal-lobbying/clients/summary?cycle=2024&id=D000029147

5. Rezzano, S., Fallini, F., & Brun, A. (2021). *DEFINING LUXURY CARS: A COMPREHENSIVE STUDY OF CRITICAL SUCCESS FACTORS AND MANAGERIAL STRATEGIES*. From https://www.researchgate.net/publication/357578050_DEFINING_LUXURY_CARS_A_COMPREHENSIVE_STUDY_OF_CRITICAL_SUCCESS_FACTORS_AND_MANAGERIAL_STRATEGIES

6. Ruiz, P., Aragón-Jurado, J. M., Seredynski, M., Cabrera, J. F., Peña, D., de la Torre, J. C., ... & Dorronsoro, B. (2023). Optimal battery management strategies for plug-in electric hybrid buses on routes including green corridors. *Sustainable Cities and Society*. From https://www.sciencedirect.com/science/article/pii/S2210670723001671#bib1